DIT047 Requirements Engineering
Final Exam

June 10, 2025

Examiner/Contact Person

Jennifer Horkoff
Jennifer will come to the room to check for questions at roughly 9 and 10:30.
Alternatively Phone: 0733 050517 (call again if no answer, I am in another exam room)

Authorized Aids

Writing instruments (pens, pencils), English Dictionary. No textbook or notes.

Grading Scale for Exams

% Grade	**Final Grade**
0-49	Fail (U)
50-64	3
65-79	4
80-100	5

PLEASE OBSERVE THE FOLLOWING:

- All answers must be in English. Answers must be legible and readable.
- Sort the questions in order before handing them in.
- Put the number of the question on every page.
- This exam has 11 pages.

Part 1: Multiple item and Short-Answer Questions

Question 1: Multiple Choice Questions (20 points)

On your paper, write the name of the question, then the letter(s) of your chosen answer(s), e.g., 1.1: c., 1.3: a, d, e. For each question, there may be more than one correct answer. For full marks, list all the correct answers. Each correct answer listed is +1 point, each incorrect answer is -1 point.

1. Which of these requirements captures design information and is therefore not well-formed? Note that these are intended to be requirements from users, not constraints on the design. The requirements are about an application which must show and process files created by another tool. In this question there may be more than one correct answer, list them all.

   (a) The icons and tool associations will be stored in a MySQL database
   (b) Each external file type should be represented as a predefined icon on the user's display, e.g., all file of certain types should have the same icon.
   (c) The user should be provided with facilities to define the type of external files, i.e., to associate tools with files
   (d) The icons used for external files must use a shade of blue
   (e) When a user select an icon representing an external file, they system shall display the warning "Opening file with external tool" in a pop-up message with options "OK" and "Cancel".
   (f) The software must provide a means of representing and accessing external files created by other tools

   Answer: a, d, e

2. Which of the following statements is correct? One or more may be correct.

   (a) Domain properties are requirements, they are requirements that come from the domain.
   (b) Constraints are not requirements, they are properties of the domain.
   (c) Constraints are requirements, they are requirements that do not come from user needs.
   (d) Users are stakeholders, not all stakeholders are users.
   (e) Stakeholders are users, not all users are stakeholders.

   Answer: c, d

3. Which of the following definitions is correct? One or more may be correct.

   (a) Transformational creativity involves taking existing ideas and converging to an agreed upon set of new ideas
   (b) Exploratory creativity involves finding new ideas by examining the combination of existing ideas
   (c) Exploratory creativity involves finding new ideas within the same space
   (d) Convergent creativity involves taking existing ideas and converging to an agreed upon set of new ideas
   (e) Transformational creativity involves shifting the idea space (e.g., by questioning assumptions), leading to a new space of ideas
   (f) Convergent creativity involves finding new ideas within the same space

Answer: c, d, e

4. Which of the following are desired qualities for user stories according to the INVEST acronym? One or more may be correct.
   (a) Easy
   (b) Estimable
   (c) Interesting
   (d) Independent
   (e) Valuable
   (f) Verifiable

   Answer: b, d, e

5. Which of the following statements about the models/diagrams we have covered in the course are true? There may be more than one correct answer.
   (a) Goal models are used to understand actors and dependencies between actors
   (b) Customer journey maps and are used to understand qualities and contributions
   (c) Use case models are used to understand actors and inputs and outputs
   (d) Context diagrams are used to understand actors and inputs and outputs
   (e) Goal models are used to understand qualities and contributions
   (f) Use case models are used to explore scenarios
   (g) Customer journey maps and are used to explore scenarios

   Answer: a, d, e, g

6. Which of the following statements about prototyping are true? There may be more than one correct answer.
   (a) In evolutionary prototypes, prototypes are thrown away
   (b) Paper prototypes are throw-away prototypes
   (c) Digital prototypes can be evolutionary prototypes
   (d) Digital prototypes can be throw-away prototypes
   (e) Paper prototypes can be evolutionary prototypes
   (f) Web-based prototypes can be evolutionary prototypes
   (g) When using paper prototypes, there is a danger that the user thinks the system is complete.
   (h) When using digital prototypes, there is a danger that the user thinks the system is complete.

   Answer: b, c, d, e, h

Question 2: Short-Answer Questions (20 points)

1. List and explain three reasons why requirements elicitation is difficult. (6 marks)

   Answer:

   Any of these well-explained will count:

   • Thin spread of domain knowledge – It is rarely available in an explicit form (i.e. not written down) – . . . distributed across many sources – . . . with conflicts between knowledge from different sources

   • Tacit knowledge (The "say-do" problem) – People find it hard to describe knowledge they regularly use – Limited Observability – The problem owners might be too busy coping with the current system – Presence of an observer may change the problem

   • Bias – People may not be free to tell you what you need to know – People may not want to tell you what you need to know

   • The outcome will affect them, so they may try to influence you (hidden agendas)

   • Stakeholder identification – Hard to figure out who to talk to

2. The following are a list of requirements. Using the desired qualities of SRS requirements, list three quality issues with one or more of these requirements. For each issue list: which requirement(s) the quality issue applies to, what is the issue, and where in the requirement it can be found. (6 points)

   Example Requirements:

   QR1 The system shall provide a payment process which is seamless from the perspective of the user.

   FR1 The system shall allow the user to scan receipts using their iPhone in order to enter ingredients into the system and the system shall allow the user to enter ingredients in the fridge manually.

   FR2 The system shall predict what recipes to suggest based on the user's mood.

   Answer: QR1: vague, not measurable FR1: not implementation free, has design with iPhone FR1: singular, this should be split FR2: feasible, how can it do this?

3. Name two methods to prioritize requirements as described in the course lectures. Describe briefly how they work. (4 marks)

   Answer:

   $100 method: ask users how much they would pay for a user story/feature, they only have $100 to spend in total. Order result by the amount spent, this is the prioritized list of requirements.

   Value vs. Cost mapping: with Value on one axis and Cost on the other, place possible user stories/features on a graph depending on their cost and value. Requirement with high value and low cost have the highest priority. Use the graph to find the ordered list of requirements.

   AHP: Analytical hierarchy process. Ask users to choose between certain features/requirements in a series of pairwise comparisons. Put these values in a table and from this a quantified priority of all features/requirements can be derived.

4. In the guest lectures you saw various examples of how different types of AI could be used to improves activities in requirements engineering. Give two examples of how AI can be used to help requirements engineering activities. Be as specific as possible. Describe one benefit and one drawback or either or both of these examples. (4 points)

   Answer: From Farnaz's lecture:

AI can help in requirements elicitation by analyzing stakeholder documentation to extract requirements or helpful information, or analyzing interview or survey text.

AI can help in requirements analysis by detecting ambiguous or inconsistent requirements.

AI can help with requirements priortization by using data from past projects to estimate priorities of requirements.

AI can help with requirements traceability by automatically finding trace links from or to or between requirements.

AI help help in change management by predicting evolving or changing requirements, or finding what requirements or code or tests are affected by changing requirements.

AI can help in regulatory compliance by seeing which requirements must be added, removed or changed when a regulation changes.

Benefits:

usually faster, reduces human effort

sometimes the AI will find interesting outputs that humans will miss

output could be more complete

make requirements engineers more competitive in industry, easier to get jobs

makes good people even better

Drawbacks:

AI can hallucinate

Some parts of the output will not be good, will have to be rejected or changed

AI is expensive, has environmental costs

A lot of AI, e.g., chatGPT, breaks privacy laws by sharing data with model owners

AI can be hard to set up, depending on the type

Gen AI requires careful prompt design, the output changes a lot depending on the prompts

If we use AI too much, human skill can be lost

Part 2: Domain Example and Long-Answer Questions

The remaining questions on the exam will relate to a problem in a domain, as described below. The scenario describes the situation today and the expectations for the new system. Focus your analysis and modeling on the envisioned (to be) system, but keep in mind the problems and requests with the as is situation, trying to avoid problems and satisfy user needs.

Teacher training at a School

Imagine that you work for the supplier of a system which manages course enrollment. Your job is to perform requirements analysis for a new system for planning and recording of courses taken by the individual employees, in particular at schools.

The situation today

There are several groups of users, for instance teachers, principals (head teacher), and school administrative staff. Users may have a dedicated desk with a computer or may move around from classroom to classroom, sharing computers or tablets. A school has both a principal and an HR manager, who are responsible for training of the staff.

All courses are run by outsourced training agencies, which often come to the school to give coursers or provide courses in their offices. New courses are announced in the monthly school news by school administrators. Some new courses are mandatory for all teachers to take within a certain period of time. When a teacher has taken a course, the principal has to record it. This is then reported to HR via an email, which goes on the teacher's employment record. They easily forget, or they record that the teacher has completed the course, when in fact they stayed at home because of illness.

The current way of keeping track of course enrollments provides a poor overview of enrollments and is hard to use. The principal and HR maintains a list of all the teachers. The list shows the courses taken by the teachers and when they were taken. It is also hard to see who needs a course in the near future.

Vision (system to be)

With the new system, the teachers should be able to enroll for courses online, and it must be easy and attractive. The enrollment is always preliminary, because the principal has to accept it. Courses are expensive, so not all enrollments will be approved.

Every month, the principal plans who should enroll in courses. For this purpose, it must be possible to see who needs the course in the near future. In order to check that a teacher has the necessary preconditions, it must be possible to see which courses the teacher has completed already.

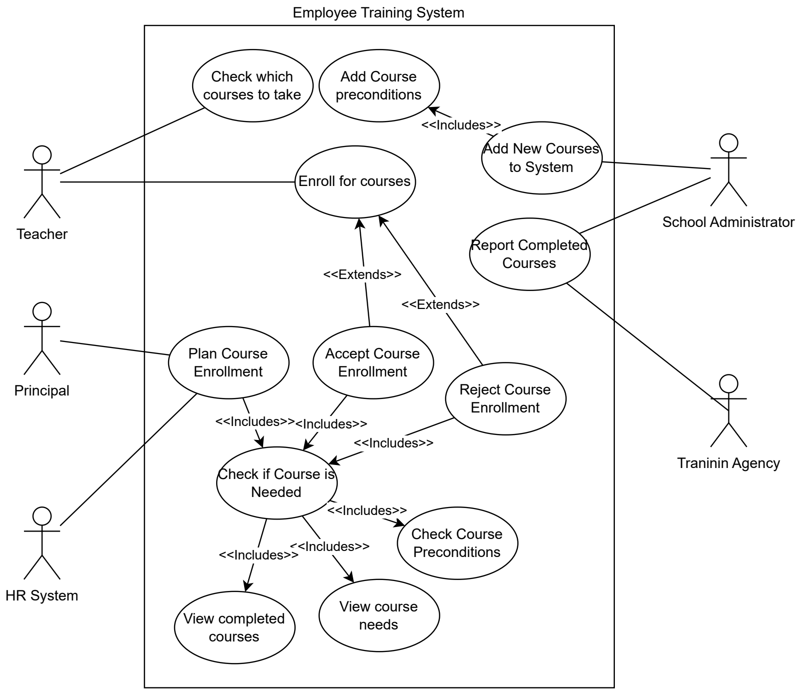
The principal must be able to record in the system which teachers should take which courses in the near future. This must also be recorded in the HR system. The teachers can then get information about in which courses they should enroll in the system. The training agencies reports back who has completed the courses, and this information is entered into the system by school administrators. School administrators still announce new courses in the school news, but also add them to the system so that teachers see them and enroll this way. Course information in the system includes necessary preconditions for taking the course.

The new system should be available in the next 18 months. It must work on the computers currently available in the school and provide privacy, including following GDPR.

Question 3: Use Case Diagram (10 points)

For the given case case, draw a use case diagram. Capture the system and other relevant stakeholders. Identify the major use cases and relationships, both between the stakeholders and the use cases, and between use cases. Try to capture three actors, and appropriate use cases. Try to use at least one example each of extends and includes in the model.

Sample Answer:
There are many possible correct variations of this diagram.
System boundary and name: 1 point
Actors: 3 points
Use cases: at least 5 with correct associations
Extends/includes: 1 point

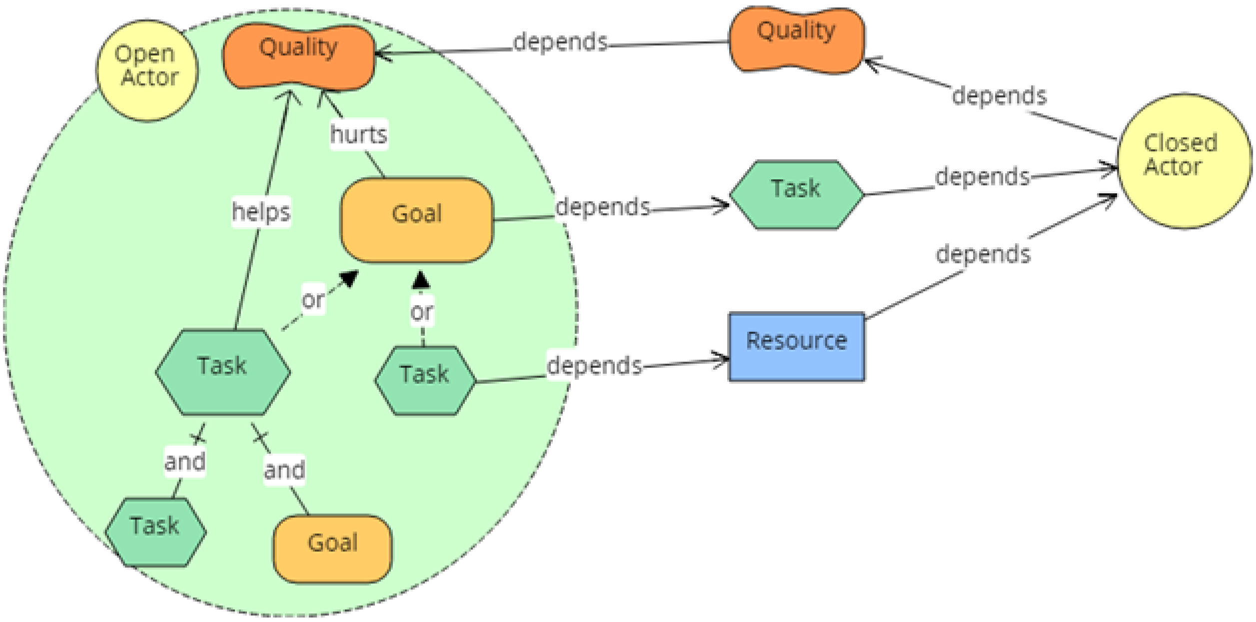


Question 4: Goal Model (10 points)

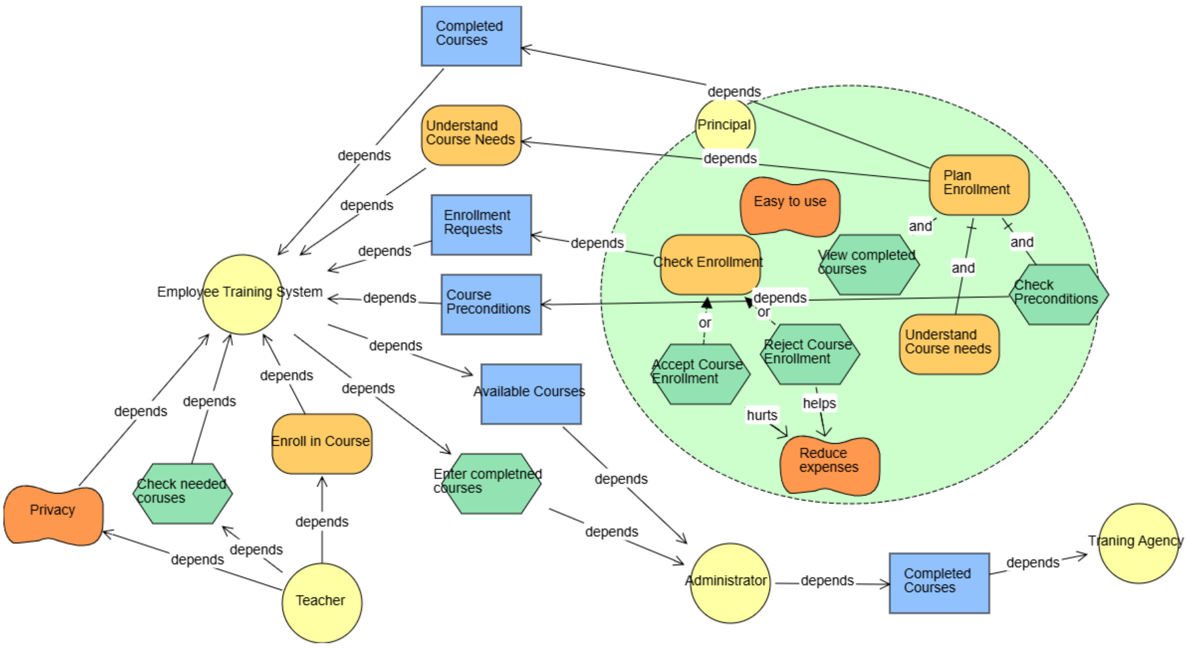
For the given case, draw a goal model. Identify the relevant actors, including the system actor. Try to capture at least four actors, and important dependencies between actors. Show the internal goals/tasks/resources/qualities for at least one actor, the others can be "closed" showing only dependencies to and from the actors. See the legend below for open and closed actors. For the open actor, capture the desired goals, tasks, resources, and qualities. Add some internal relationships between these elements.

There are many elements that can be included in the model. Aim for: four actors including one open actor with an actor boundary, at least one goal, quality, resource, task, four dependency links, two contribution links and two AND/OR links. In other words, the model does not have to be complete as per the domain description. It's helpful to use the information already captured in the other diagrams; however, we won't mark you on consistency between diagrams. Supplement the diagram with text to explain any ambiguous or unclear parts of the model.

Goal Model Legend (note, can use alternative shapes and labels as long as the mapping to the original concepts is clear)



Sample Answer:
Actors: 2 marks
Goals: 1 mark
Qualities: 1 mark
Tasks: 1 mark
Resources: 1 mark
Dependencies: 2 marks
Contributions: 1 mark
AND/OR: 1 mark



Question 5: Customer Journey Map (10 points)

Draw a customer journey map for the given case. Indicate the persona that the journey applies to. Think about the channels and touchpoints that the persona would have to go through when using the system. Try to include three channels and at least five touchpoints.

Sample Answer:

Persona: 1 mark
Channels: 3 marks
Touchpoints: 5 marks
Timeline: 1 mark
Here are two possible answers

Persona: Teacher, Jim, Age 42

	See New Courses	Check needed courses	Enroll in Courses	See course approved	Take Course
Newsletter	★				
Enrollment System on Classroom Computer		★	★		
Enrollment System on shared Tablet				★	
School					★

Persona: School Administrator, Janice, Age 55

	Get List of New Courses	Announced courses	Enter courses	Enter Preconditions	Get Course completions	Enter Completions
Training Agency Email (or call)	★				★	
Newsletter		★				
Enrollment System			★	★		★

Question 6: Textual Requirements (20 points)

Write SRS requirements and user stories for the case provided. List four functional requirements in SRS form, two non-functional requirements in SRS form, two domain assumptions, and two constraints. In addition, list five user stories, each with acceptance criteria. The user stories should be unique, i.e. not a repeat of one of the SRS requirements. Remember the desired characteristics of user stories and SRS requirements when writing your requirements.

Sample Answers:

Functional Requirements

1. The system shall allow users to request enrollment in available courses.

2. The system shall allow users with assigned permissions to reject or accept enrollment requests.

3. The system shall allow users with assigned permissions to view all courses completed by all users.

4. The system shall allow users with default permissions to view the courses that they have completed.

5. The system shall allow entering of new courses into the system, including course preconditions.

NFRs 6. The system shall be easy to use, user testing should be conducted with at least five potential users and at least four users should report the system is easy to use.

7. The system should preserve privacy, if a user is inactive for five minutes, they should be logged out.

8. It should be easy to enroll in a course, four out of five test users should be able to enroll within one minute.

Domain Assumptions

9. We assume that everyone in the school has access to the system.

10. We assume that the training agency and the school administrator have a way to communicate courses and enrollment.

11. We assume the training agencies accurately report course completions.

Constraints

12. The system must be complete in December 2026.

13. The system must follow all rules as part of the GDPR regulation.

User Stories

15. As a principal I want to be able to plan needed course enrollments, so the teachers take the courses they need in time to follow the rules.

Acceptance criteria:

Principals can plan course enrollment for all teachers monthly in less than 3 hours.

At the end of course enrollment, all teachers have been considered.

All rules about teachers taking courses are followed if the plan is enacted.

16. As a principal, when planning course enrollments, I need to be able to view course preconditions, so I can see if teachers have the necessary preconditions.

Acceptance criteria:

All preconditions for each course are visible and correct.

Principals can view all courses that each teacher has taken.

Course names and numbers are consistent.

17. As a teacher, I want to be able to see what courses I need to take, so I can enroll in the courses.

Acceptance criteria:

The teacher can see a list of courses which the principal has indicated they need to take.

If there are no courses to take, a message tells the teacher there are no courses.

18. As a school administrator, I want to be able to enter courses the teachers have completed, so it goes on their record.

Acceptance criteria:

Completed courses entered can now be viewed by teachers and administrators.

Administrators should be able to enter multiple course completions for a course at once, to reduce time. Entering all teachers that complete a course should take less than 5 minutes.

19. As a principal, I want to be able to get a list of which teachers need which courses, so I can record them in the HR system.

Acceptance criteria:

It should be possible to get a summary list of courses that are needed by which teachers that are new in the last month, since the process of course planning was last performed.

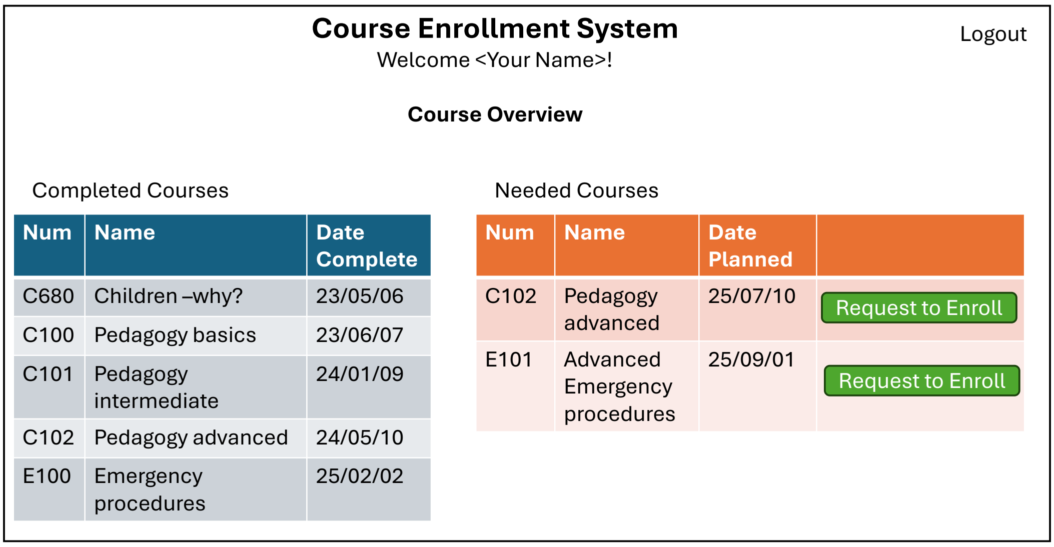
This list should be easy to read, so it can be recorded into the HR system in less than 15 minutes.

Question 7: Prototype (10 points)

Draw one prototype screen for the given case implementation. The screen can include pop-up windows, just be clear how and why they appear. The purpose of the screen should be clear. The screen should implement at least three functional requirements (SRS or user stories). These requirements should be clearly implemented on this screen. The requirements do not have to be the requirements from previous questions. Write the full text of the two requirements implemented here, even if they are not new, so we do not have to flip back to find them. For the screen, write: the three requirements implemented, and explain how and exactly where the requirements are implemented.

Sample Answers:

We give 6 points for the requirements,
and 4 points for the screen.



Implemented Functional Requirements

1. The system shall allow users to request enrollment in available courses.

how and where: The buttons on the far right of the screen associated with each needed course allows teachers to request enrollment for each course.

4. The system shall allow users with default permissions to view the courses that they have completed.

How and where: This is shown in the table on the left under Completed courses after the teacher has logged in.

17. As a teacher, I want to be able to see what courses I need to take, so I can enroll in the courses.

How and where: This is shown by the table of needed courses on the right of the screen.